Optimizing Network Topology for Cascade Resilience

Alexander Gutfraind

Abstract Complex networks need resilience to cascades - to prevent the failure of a single node from causing a far-reaching domino effect. Such resilience can be achieved actively or topologically. In active resilience schemes, sensors detect the cascade and trigger responses that protect the network against further damage. In topological resilience schemes, the network's connectivity alone provides resilience by dissipating nascent cascades. Designing topologically resilient networks is a multi-objective discrete optimization problem, where the objectives include resisting cascades and efficiently performing a mission. Remarkably, terrorist networks and other "dark networks" have already discovered how to design such networks. While topological resilience is more robust than active resilience, it should not always be pursued because in some situations it requires excessive loss of network efficiency.

Key words: complex networks, cascade resilience, multi-objective optimization, epidemics on networks, terrorist networks

1 Introduction

Cascades are ubiquitous in complex networks and they have inspired much research in modeling, prediction and mitigation [63, 20, 14, 36, 11, 58, 61, 55, 73, 54]. For example, since many infectious diseases spread over contact networks a single carrier might infect other individuals with whom she interacts. The infection might then propagate widely through the network, leading to an epidemic. Even if no lives are lost, recovery may require both prolonged hospitalizations and expensive treatments. Similar cascade phenomena are found in other domains such as

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power distribution systems [22, 44, 39], computer networks such as ad-hoc wireless networks [61], financial markets [8, 37] and socio-economic systems [41]. A particularly interesting class are "dark" or clandestine social networks, such as terrorist networks, guerrilla groups [66], espionage and crime rings [5, 53]. In such networks if one of the nodes (i.e. individuals) is captured by law enforcement agencies, he may betray all the nodes connected to him leading to their likely capture.

Dark networks are therefore designed to operate in conditions of intense cascade pressure. As such they can serve as useful prototypes of networks that are cascade-resilient because of their connectivity structure (topology) alone. Their nodes are often placed in well-defined cells - closely-connected subnetworks with only sparse connections to the outside (for an example from World War II see Fig. 1) [52]. The advantages of cells are thought to be that the risk from the capture of any person is mostly limited to his or her cell mates, thereby protecting the rest of the network [29, 49]. Modern terrorist groups retain this cellular structure, but increasingly use networks made of components with no connections between them, thus caging cascades within each component [68, 70, 74].

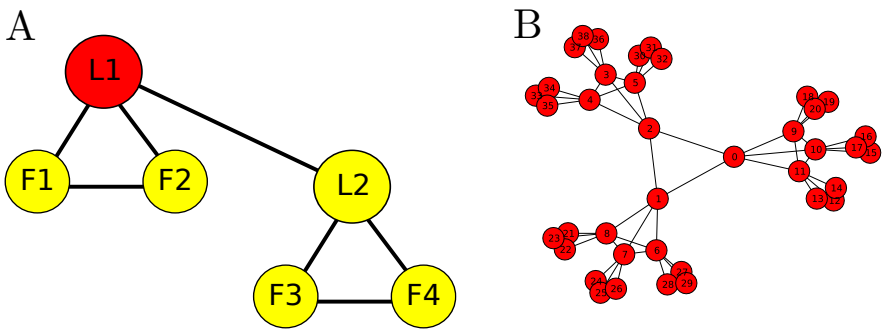


Fig. 1 The French World-War II underground network *Francs-tireurs et Partisans* (FTP) reconstructed by the author based on the account in [52]. Its organizational unit was the combat group (A). In an idealized case, nor always followed, this was divided into two "teams" of three fighters, where leader L1 was in overall command and in command of team 1. His lieutenant, L2, led team 2 and assumed overall command if L1 was captured. The small degree of the nodes ensured that the capture of any one node did not risk the exposure of a significant fraction of the organization. Each "group" is in a command hierarchy (B) where 3 groups (bottom-level nodes) made a "section", 3 sections made a "company", and finally 3 companies made a "battalion".

Active versus Topological Cascade Resilience

Networks could be endowed with cascade resilience using two complementary approaches: "active" and "topological". In active resilience, the network is monitored for cascades, and if a cascade is detected, attempts are made to stop it *while it progresses*. For example, in case of human pathogens, health authorities may continuously monitor hospital records for contagious diseases. If the records begin to

show anomalous increases, various responses are initiated, including distribution of medicines and alerts to the the public.Similarly, in power distribution systems, special devices monitor the network for signs of cascades, such as high currents or phase changes. Those may indicate failures in lines, short circuits and other phenomena that threaten to disrupt the system or damage its components. The power system includes a variety of automated controls tasked with stopping the nascent cascade [54], such as "relays". Those relays can automatically shut down faulty lines or nodes of the network so as to isolate them from the rest of the network [64, 76].

Those two examples of active cascade resilience must be contrasted with "topological" approach to resilience where only the topology (i.e. the pattern of connections) is used to increase cascade resilience. For example, the network could be structured into modules, where two modules are connected to each other through long paths. As a result, in certain types of cascades, a failure in one module might dissipate before it reaches any other module. When topological cascade resilience is possible, it offers two advantages over active resilience: simplicity and robustness. In topological resilience the network protects itself, requiring no real-time automated decisions or difficult-to-achieve rapid response during the cascade.

2 What is a Cascade-Resilient Network?

The words Network, Cascade and Resilience have many domains of application, so much so that no universal definition of those terms exists. Therefore, this section briefly surveys some of the recurrent applications and definitions of those terms. It also introduces specific definitions for those terms - definitions that are appropriate for some applications. Later in the paper those definitions serve as an example of optimizing networks for cascade resilience.

2.1 Network as an Unweighted Graph

Complex Networks is the study of real world systems using ideas of graph theory. Specifically, here and in most other studies the network is represented using simple unweighted graph G: a tuple (V,E) where V is a set called "nodes" or "vertices" and E are two-element subsets of V termed "edges". Such an approach offers simplicity and can employ the well-developed tools of graph theory. Ultimately through, models of networks must consider their evolving nature, fuzzy boundaries and multiplicities of node classes and diverse relationships.

The simplification is often unavoidable given the lack of data on networks. For example, in dark social networks only the connectivity is known, if that. Fortunately, the loss of information involved in representing networks as simple rather than as weighted graphs could be evaluated. Appendix 1 examines two unusually rich data sets (two dark networks) where the edges could be assigned weights. It is shown

that for those networks the error in using simple graphs has no systematic bias and is usually small.

2.2 Cascades

There is a very extensive literature on both cascades and resilience. The classic literature on cascades includes two basic models: percolation cascades and capacity cascades. The former originate in Physics but are often applied to Epidemiology, where they are termed "contagions" or "epidemics" (see e.g. [59]). In percolation phenomena, nodes are assigned states which change because of the influence of their neighbors. For example, an infected node can pass the infection to its contacts in the network, and the infection could then be passed to more and more nodes. One variant of such percolations are the case where nodes change their state only when a certain fraction of their neighbors exert influence (see e.g. [73, 14]). Percolation phenomena are exceptionally well-studied, and in many variants analytic expressions exist for the final extent of the cascade as a function of the network topology (see e.g. [58, 26]). The capacity cascades are characteristic of capacitated networks, such as power transmission systems and supply chains. Classically, in those systems the edges are assigned capacities and thus carry flows from supply nodes to demand nodes. Cascades occur when due to failures the flow can no longer be carried by the edges within their capacities or when some of the supply nodes fail [21, 44, 54, 3]. In capacity cascades the failure can jump to nodes that are many hops away from the initial failures possibly skipping the neighbors.

2.3 Resilience

A vast number of studies attempted to define resilience, often in very different ways. Perhaps the most common meaning refers to the connectivity of the network under disruption or failures in its components. Such definitions are motivated by applications in telecommunications where it is desired that nodes are able to find a path to each other even if some of the components in the networks are damaged or destroyed [6, 16, 7, 1, 12, 15, 23, 35, 34, 42].

The idea of damaging networks has attracted a lot of research in the area of Sociology of secret societies such as terrorist networks [29, 75, 33, 25, 53, 4, 50, 49]. In fact many secret societies are benign, including non-governmental organizations and dissident movements operating in hostile political environments. In those networks, if the network is "penetrated" by its enemy, it must be able to minimize the damage. Economists recently analyzed the problem of organizational design when the organization is being attacked [27]. Related problems have also been studied in Epidemiology, where the question focused on immunization strategies (e.g. [65]) but apparently not as a question of optimal network design.

Recently, resilience has became associated with the ability to quickly recover from damage rather than to absorb it [10, 32]. Indeed, in many applications disruptions and failures are not rare singular events, but rather occur regularly and even continuously. For example, the are continuous demand spikes in communication networks [19] and voltage fluctuations in power systems.

It is to be expected that no notion of resilience would be useful universally across different applications. Similarly, many networks experience cascades, but the details vary. This paper will investigate cascade resilience under a particularly important and well-characterized class of cascades known as "SIR" (susceptible-infected-recovered.) SIR are a type of percolation where any failed node leads to the failure of each neighboring node independently with probability τ [59]. This τ represents the network's propensity to experience cascades, expressing both the susceptibility of components and the environment in which the network operates. Using the SIR model, resilience $R(G)$ could be defined as the average fraction of the network that does not fail in the cascade:

$$R(G) = 1 - \frac{1}{n-1}\mathbb{E}[\text{extent of a cascade}]\,, \tag{1}$$

where "extent of a cascade" refers to the ultimate number of new cases created by a single failed node (the initial node does not count.) For simplicity, cascades are assumed to start at all nodes with uniform probability.

Observe that under this definition the most cascade-resilient network ($R(G) = 1$) is the network with no edges. But such a network cannot carry any information from node to node! It is not surprising that the objective of designing cascade resilience conflicts with other features of the network. In other cascade types, such as cascades on capacitated networks, the most cascade-resilient network is the network with infinite capacities, which obviously would conflict with the objective of minimizing cost. It follows then, that optimization of the networks requires specifying a notion of value or efficiency.

2.4 Measuring Efficiency

Notions of network efficiency attempt to quantify the value of a network, and this problem has a long history. For example, an influential early work on communication networks suggested that a network's value increases as $O\left(\frac{n(n-1)}{2}\right)$, where $n = |V|$ is the number of nodes, because each node can connect to $n-1$ other nodes [9]. However, this measure ignores the difficulty of connecting to other nodes (as well as, e.g. cost). Indeed, it is often desired that the distances between the nodes are short: when nodes are separated by short distances they can, for example, more easily communicate and distribute resources to each other. Therefore, many authors invoke measures based on the distances between pairs of nodes in the network (see e.g. [45, 56, 50].)

In the following we will consider a version of distance-based efficiency, termed "distance-attenuated reach" metric [45]. For all pairs of nodes $u, v \in V$, weigh each pair by the inverse of its internal distance (the number of edges in the shortest path from u to v) taken to power g:

$$W(G) = \frac{1}{n(n-1)} \sum_{u \in V} \sum_{v \in V \setminus \{u\}} \frac{1}{d(u,v)^g}, \tag{2}$$

Normalization by $n(n-1)$ ensures that $0 \leq W(G) \leq 1$, and only the complete graph achieves 1. As usual, for any node v with no path to u, set $\frac{1}{d(u,v)^g} = 0$. The parameter g, "connectivity attenuation" represents the rate at which distance decreases the connectivity between nodes. Unless stated otherwise $g = 1$. A valuable property of $W(G)$ is that it is well-defined and non-singular even on networks that have multiple components with no connections to each other. As will be shown, such a separation into components provides a very powerful mechanism for cascade resilience.

3 Evaluating Real Networks

Significant insight into cascade resilience can be derived from comparing the cascade resilience of networks from different domains. We will see that dark networks like terrorist networks are more successful in the presence of certain cascades than other complex networks. Their success stems not from cascade resilience alone but from balancing resilience with efficiency.

To make those comparisons, define the overall "fitness", $F(G)$, of a network by aggregating resilience and efficiency through a weight parameter r:

$$F(G) = rR(G) + (1-r)W(G).$$

The parameter r depends on the application and represents the damage from a cascade - from light ($r \to 0$) to catastrophic ($r \to 1$). It is possible to include in fitness other metrics such as construction cost.

We will compare the fitnesses of several complex networks, including communication, infrastructure and scientific networks to the fitnesses of dark networks. The class of dark networks will be represented by three networks: the 9/11, 11M and FTP networks. The 9/11 network links the group of individuals who were directly involved in the September 11, 2001 attacks on New York and Washington, DC [43]. Similarly the 11M network links those responsible for the March 11, 2004 train attacks in Madrid [68]. Both 9/11 and 11M were constructed from press reports of the attacks. Edges in those networks connect two individuals who worked with each other in the plots [43, 68]. The FTP network is an underground group from World War II (Fig. 1), whose network was constructed by the author from a historical account [52].

Figure 2 shows that the dark networks attain the highest fitness values of all networks, except for extreme levels of cascade risk ($\tau > 0.6$) (Appendix 2 shows the full efficiency and resilience values.) This is to be expected: only 11M, 9/11, and the FTP networks have been designed with cascade resilience as a significant criterion - a property that makes them useful case studies. For high cascade risks ($\tau > 0.6$) the CollabNet network exceeds the fitnesses of the dark networks. CollabNet was drawn by linking scientists who co-authored a paper in the area of network science [60]. It achieved high fitness because it is partitioned into research groups that have no publications with outside scientists. Like some terrorist networks, it is separated into entirely disconnected cells.

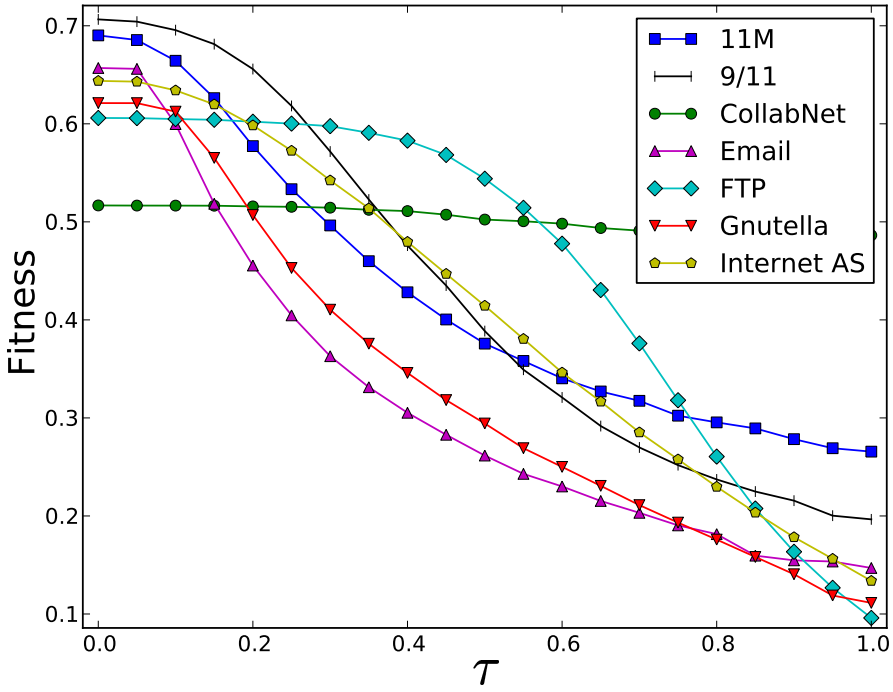


Fig. 2 Fitnesses of various networks at $r = 0.51$ and various values of τ. 11M is the network responsible for the March 11, 2004 attacks in Madrid (70 nodes, 240 edges.) 9/11 [43] is the network responsible for the 9/11 attacks (62 nodes, 152 edges.) CollabNet [60] is a scientific co-authorship network in the area of network science (1589 nodes, 2742 edges.) E-Mail [28] is a university's e-mail contact network, showing its organizational structure (1133 nodes, 5452 edges.) FTP is the network in Fig. 1 (174 nodes, 300 edges.) Gnutella [67, 38] is a snapshot of the peer-to-peer network (6301 nodes, 20777 edges.) Internet AS [48] is a snapshot of the Internet at the autonomous system level (26475 nodes, 53381 edges.) Except for $\tau > 0.6$ dark networks (11M, 9/11 and FTP) attain the highest fitness.

The 9/11 and the 11M networks are very successful for low values of τ (< 0.2), but then rapidly deteriorate because of a jump in the extent of cascades - the so-called percolation transition [24]. Past this threshold, cascades start affecting a large

fraction of the network, resilience collapses and the fitness declines rapidly. The pattern of onset of failure can be clearly seen in most of the networks. For violent secret societies this transition means that the network might be initially hard to defeat, but there is a point after which efforts against it start to pay off. Because τ is representative of the security environment, the 9/11 network is found to be relatively ill-adapted to the more stringent security regime implemented after the attacks. Indeed, it is likely that the 9/11 attacks would have been thwarted under the current security regime since some of the nodes were captured before the attacks, but not interrogated in time to discover and apprehend the rest of the network [72]. In contrast, the cellular tree hierarchy of the FTP network is more suitable for an intermediate range of cascade risks. However, the pair-wise distances in it are too long to provide high efficiency. Therefore, its fitness is comparatively poor in the very low and very high values of τ.

4 Designing Networks

The success of dark networks must be due to structural elements of those networks, such as cells. If identified, those elements could be used to design more resilient networks and to upgrade existing ones. Thus, by learning how dark networks organize, it will be possible to make networks such as communication systems, financial networks, and others more resilient and efficient.

Those identification and design problems are our next task. We propose to solve both using an approach based on discrete optimization. Let a set of graphs $\mathbb{G}$ be called a "network design" if all the networks in it share a structural element. Since dark networks are often based on dense cliques, we consider a design where all the networks consist of one or multiple cliques. We consider also designs based on star-like cells, cycle-based cells and more complex patterns (see Fig. 3 and Appendix 3 for the exact set of networks.)

In the first step we will find the most successful network within each design. Namely, consider an optimization problem where the decision variable is the topology G of a simple graph taken from a design $\mathbb{G}$. The objective is the fitness $F(G)$:

$$\max_{G \in \mathbb{G}} F(G)\,. \tag{3}$$

In the second step (next section) we will compare the fitnesses across designs, thus identifying the topological feature with the highest fitness (e.g. star vs. clique.)

This optimization problem introduces a method for designing cascade-resilient networks for applications such as vital infrastructure networks. To apply this to a given application, one must make the design $\mathbb{G}$ the set of all feasible networks in that domain, to the extent possible by computational constraints. In the area of terrorist networks, the model is closely related to the game-theoretic work of Lindelauf et al. [50, 49].

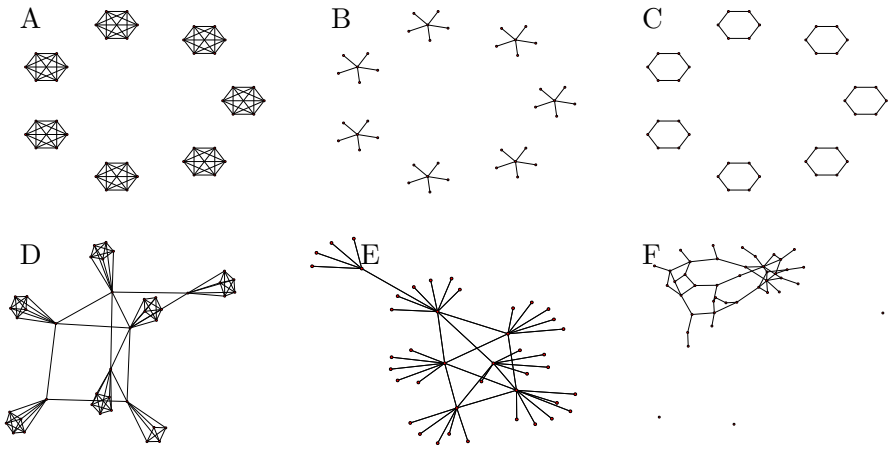


Fig. 3 Graphs illustrating the 6 **network designs.** Cliques (A), Stars (B), Cycles (C), Connected Cliques (D), Connected Stars (E), and Erdos-Renyi "ER" (F). Each design is configured by just one or two parameters (the number of individuals per cell and/or the random connectivity.) This enables rapid solution of the optimization problem. In computations the networks were larger ($n = 180$ nodes.)

Multi-Objective Optimization Approach

A complementary approach is to consider the multi-objective optimization problem in which $R(G)$ and $W(G)$ are maximized simultaneously:

$$\max_{G \in \mathbb{G}} \{R(G), W(G)\} \, . \tag{4}$$

The multi-objective approach cannot find the optimal network but instead produces the Pareto frontier of each design - the set of network configurations that cannot be improved without sacrificing either efficiency or resilience. The decision maker can use the frontier to make the optimal trade-off between resilience and efficiency.

The fitness and the multi-objective optimization approaches could be easily generalized to consider additional design objectives and constraints. For example, research on social networks indicates that resilience and efficiency might be just two of several design criteria that also include e.g. "information-processing requirements", that impose additional constraints on network designs [5]. In the original context "information-processing" refers to the need to have ties between individuals involved in a particular task, when the task has high complexity. Each individual might have a unique set of expertise into which all the other agents must tap directly. Generalizing from sociology, such "functional constraints" might considerably limit the flexibility in constructing resilient and efficient networks. For example, in the context of terrorism, this constraint significantly decreased the quality of attacks that could be successfully carried out in the post 9/11 security environment [74]. Such functional constraints could be addressed by looking at a palette of network designs

which already incorporate such constraints. In engineering applications, such as infrastructure or communication networks, the financial cost of building the network is another key objective.

The multi-objective problem (4) and the scalarized model (3) are hard: both are discrete optimization problems with non-linear objective functions. In small networks, those could be solved using brute force enumeration. In larger networks, solutions could be obtained using derivative-free optimization methods [18] and approximations such as [69]. Promising approaches also exist for finding the Pareto front [51, 46, 57]. Whether those methods are fast and accurate enough would depend on the definitions of $R(G)$, $W(G)$ and the set $\mathbb{G}$.

5 Topological Cascade Resilience in the SIR-Reach Model

In this section and the rest of the paper we will use the SIR-Reach model (R and W follow Eqs. 1,2) The two models are attractive because they have real applications: Social Networks and Epidemiology. They have also been extensively explored by network scientists, which make them ideal as a case study in topological cascade resilience.

5.1 Optimal Network

The first set of experiments compares the designs against each other under different cascade risks (τ), Fig. 4. At each setting of τ, each design is optimized to its best configuration, i.e. the best cell size, and connectivity if applicable. The curves indicate the fitness of the optimal network in each design. Typically at each τ the optimal network is different from the optimal network at another τ. Observe that within each design, as τ increases the fitness decreases - one cannot win when fighting cascades, only delay (see Appendix 6 for the proof.) In certain applications it is possible to invest in reducing the cascade propagation probability, τ. Then the curves in Fig. 4 could also be viewed as expressing the gain from efforts to reduce cascades by reducing τ and also adapting the network structure. If the slope is steep then the gains are large.

Comparing designs to each other reveals that Connected Stars is superior to all others in fitness (Fig. 4.) The design also outperforms any of the empirical networks in Fig. 2 in part because for each value of τ we selected the optimal network. The simpler Stars design is almost as fit, deteriorating only at extreme ranges of τ. The rankings of the designs are of course dependent on the parameter values, but not strongly (see Appendix 4 for the proof.) Star-like designs are successful because the central node in a star acts as a cascade blocker while keeping the average distance in the star short (~ 2). Only for sufficiently low r, the Cliques, Connected Cliques and Connected Stars designs are superior to the Stars design. For such values of r

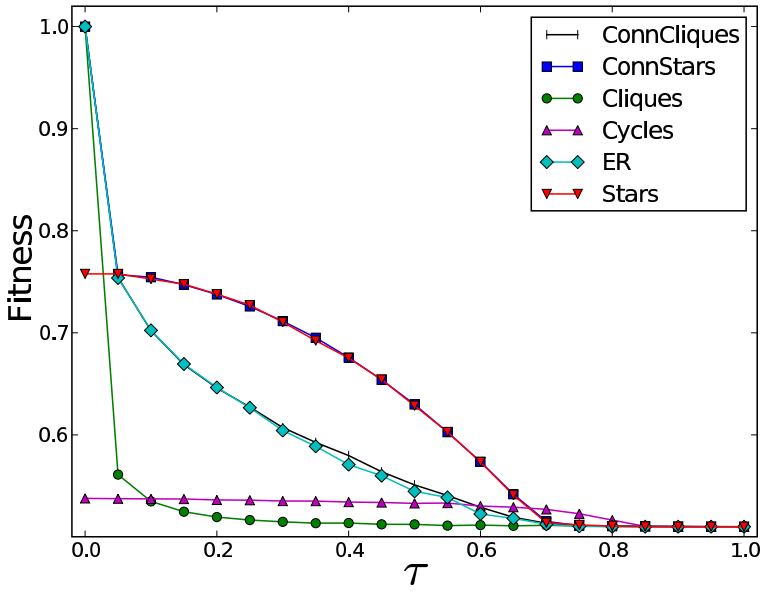


Fig. 4 Fitness at $r = 0.51$ of various network designs. The Connected Stars design is the best design at all cascade risks, τ. Cliques and Connected Cliques are competitive only for extreme ranges of τ. The superiority of Connected Stars over the ER (random graph) confirms the hypothesis that cells give fitness gains against cascades. The fitness of a design at each value of τ is defined as the fitness of the optimal configuration (network ensemble) within that design.

efficiency is the dominant contributor to fitness. High weighting for efficiency benefits the former designs where efficiency can be 1 by constructing a fully connected (complete) graph (see Appendix 5 for the details about the resilience and efficiency of the networks and Appendix 7 for analytic results.) In the star design efficiency is lower, reaching $\sim \frac{1}{2}$ (when all nodes are placed in a single large star.)

It has been long conjectured that cells provide dark networks with high resilience. Indeed, this is probably the reason why we found that dark networks have higher fitnesses than other networks. But cells also reduce the efficiency of a network since they isolate nodes from each other. To rigorously determine the net effect of cells, we compare the ER design (random graphs) to the Connected Stars design. ER is a strict subset of Connected Stars but only Connected Stars has cells. Therefore it is notable that Connected Stars has a higher fitness than ER, often significantly so. Indeed, cells must be the cause of higher fitness because cells are the only feature in Connected Stars that ER lacks.

5.2 Properties of Optimal Networks

Many properties of the optimal networks such as resilience, efficiency and edge density show rapid phase transitions as r is changed. For example, in the Cliques design when $r < 0.5$ the optimal network has high density that maximizes efficiency, whereas for $r > 0.5$ it is sparse and maximizes resilience (Fig. 5.)

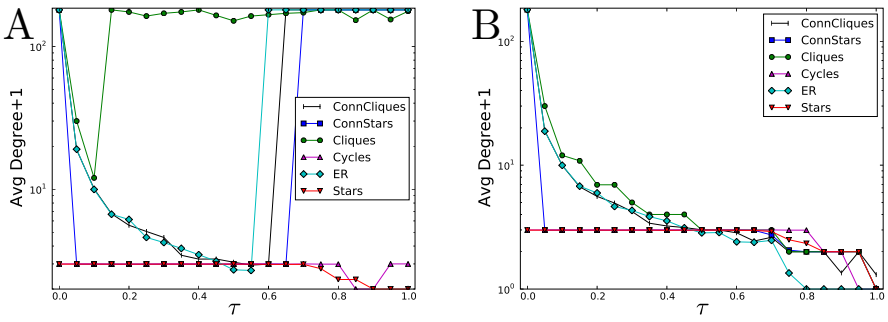


Fig. 5 Average degree in the optimal configuration of each design. At $r = 0.49$ (A) the optimization prefers networks that have high efficiency while at $r = 0.51$ (B) the preference is for resilience. In (B) the average degree diminishes monotonically to compensate for increasing cascade risk. In (A) most designs have a threshold τ at which they jump back to a completely-connected graph because structural cascade resilience becomes too expensive in terms of efficiency.

Intuition may suggest that the networks grow more sparse as cascade risk grows. Instead, the trend was non-monotonic (Fig. 5.) For $\tau \gg 0$ and $r < 0.5$ Cliques, Connected Cliques and Connected Stars became denser, instead of sparser, and for them the most sparse networks were formed in the intermediate values of τ where the optimal networks achieve both relatively high resilience and high efficiency. At higher τ values, when $r < 0.5$ it pays to sacrifice resilience because fitness is increased when efficiency is made larger through an equal or lesser sacrifice in resilience. The Stars design does not show a transition at $r = 0.5$ because it is hard to increase efficiency with this design (see details about the structure in Appendix 8 and sensitivity analysis in Appendix 9.)

5.3 Multi-objective Optimization

A complementary perspective on each design is found from its Pareto frontier of resilience and efficiency (Fig. 6.) Typically a design is dominant in a part of the Resilience-Efficiency plane but not all of it. The Stars and Connected Stars designs can access most of the high resilience-low efficiency region. In contrast, the Cliques and Connected Cliques can make networks in the medium resilience-high efficiency regions.

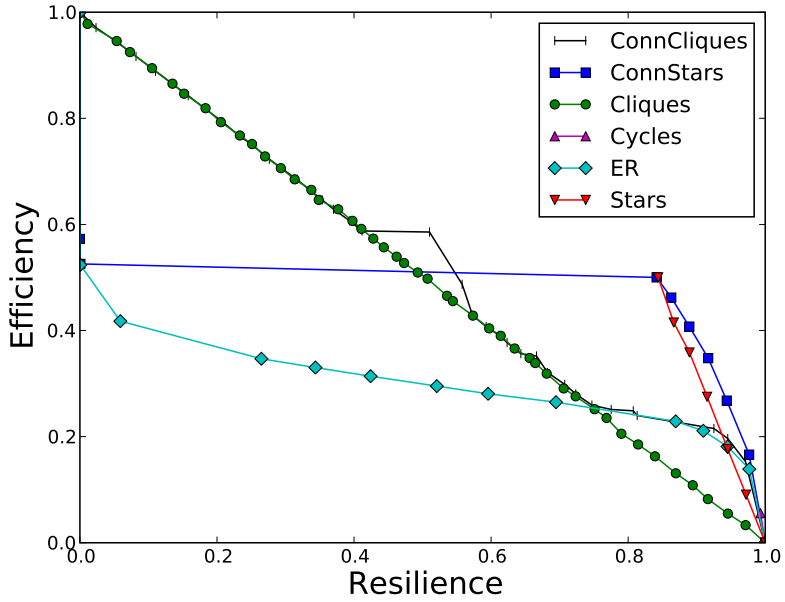


Fig. 6 The Pareto frontiers of various network designs ($\tau = 0.4$). The configurations of the Connected Stars design dominate over other designs when the network must achieve high resilience. However, designs based on cliques are dominant when high efficiency is required. Several designs show sharp transitions where at a small sacrifice of efficiency it is possible to achieve large increases in cascade resilience.

The sharp phase transitions discussed earlier are seen clearly: along most of the frontiers, if we trace a point while decreasing resilience, there is a threshold at which a small sacrifice in resilience gives a major gain of efficiency. More generally, consider the points where the frontier is smooth. By taking two nearby networks on the frontier one can define a rate of change of efficiency with respect to resilience: $\left|\frac{\Delta W}{\Delta R}\right|$. The ratio can be used to optimize the network without using the parameter r. When $\left|\frac{\Delta W}{\Delta R}\right| \gg 1$ the network optimizer should choose to reduce to the resilience of the network in order to achieve great gains in efficiency; when $\left|\frac{\Delta W}{\Delta R}\right| \ll 1$ efficiency should be sacrificed to improve resilience.

6 Discussion

The analysis above considered both empirical networks and synthetic ones. The latter were constructed to achieve structural cascade resilience and efficiency. In contrast, in many empirical networks the structure emerges through an unplanned

growth process or results from optimization to factors such as cost rather than blocking cascades. Without exception the synthetic networks showed higher fitness values despite the fact that they were based on very simple designs. This suggests that network optimization can significantly improve the fitness and cascade resilience of networks. It follows that an optimization process can be applied to design a variety of networks and to protect existing networks from cascades.

Many empirical networks also have power-law degree distributions [59]. Unfortunately, this feature significantly diminishes their cascade resilience: the resulting high-degree hubs make the networks extremely vulnerable to cascades once τ is slightly larger than 0 [63, 20].

In some successful synthetic networks the density of edges increased when the cascade risk τ was high. This phenomenon has interesting parallels in non-violent social movements which are often organized openly rather than as secret underground cells even under conditions of severe state repression [71]. This openness greatly facilitates recruitment and advocacy, justifying the additional risk to the participants, just like the sacrifice of resilience to gain higher efficiency is justified under $r < 0.5$ conditions.

There are other important applications of this work, such as the design of power distribution systems. For power networks, the definition of resilience and efficiency will need to be changed. It would also be necessary to use much broader designs and optimization under design constraints such as cost. Furthermore, this work could also be adapted to domains of increasing concern such as financial credit networks, whose structure may make them vulnerable to bankruptcies [8, 37].

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Appendix 1: Resilience and Efficiency of Weighted Networks

In some networks, each edge (u,v) carries a distance weight $D_{uv} > 0$. The smaller the distance, the closer the connection between u and v. To compute the fitness of those networks, we now introduce generalizations of resilience and efficiency. Those reduce to the original definitions for unweighted networks when $D_{uv} = 1$, while capturing the effects of weights in the weighted networks.

The original definition of resilience was built on a percolation model where the failure of any node leads to the failure of its neighbor with probability τ. In the weighted network more distant nodes should be less likely to spread the cascade. Thus we make the probability of cascade through (u,v) to be $\min(\tau/D_{uv}, 1)$.

The efficiency was originally defined as the sum of all-pairs inverse geodesic distances, normalized by the efficiency of the complete graph. In the weighted network,

both the distance and the normalization must be generalized. To compute the distance $d(u,v)$ we consider the weights on the edges D and apply Dijkstra's algorithm to find the shortest path. Normalization too must consider D because a weighted graph with sufficiently small distances could outperform the complete graph (if all the edges of the latter have $D_{ij} = 1$.) Therefore, we weigh the efficiency by the harmonic mean H of the edges (E) of the graph:

$$W(G) = \frac{H(G)}{n(n-1)} \sum_{u \in V} \sum_{v \in V \setminus \{u\}} \frac{1}{d(u,v)^g}, \tag{5}$$

where

$$H(G) = \frac{|E|}{\sum_{(u,v)\in E} \left(\frac{1}{D_{uv}}\right)^g}.$$

The harmonic mean ensures that for any D, the complete graph has $W(G) = 1$.

Having defined generalized resilience and efficiency we can evaluate the standard approach to dark network, which represents them as binary graphs $D_{uv} \in \{0,1\}$, rather than as weighted graphs. The former approach is often taken because the information about dark networks is limited and insufficient to estimate edge weights.

Fortunately, in two cases, the 9/11 network and the 11M network [43, 68] the weights could be estimated. The 9/11 data labels nodes as either facilitators or hijackers. Hijackers must train together and thus should tend to have a closer relationship. Thus set $D_{uv} = 2, 1, 0.5$ if the pair u, v includes zero, one or two hijackers, respectively. The 11M network is already weighted ($Z_{uv} = 1, 2, 3\ldots$) based on the number of functions each contact (u,v) serves (friendship, kin, joint training etc.) We mapped those weights to D by $D_{uv} = 2/Z_{uv}$. In both networks the transformation was so that the weakest ties have weight 2, giving them greater distance than in the binary network, while the strongest ties are shorter than in the binary network.

Figure 7 compares the fitnesses, resiliences and efficiencies of the weighted and binary representations. It shows that for both networks, the fitnesses of the binary representation lies within 0.15 of the fitness of the weighted representation and for some τ much closer. The efficiency measures are even more close (within 0.05.) The behavior of resilience is intriguing: for the 9/11 network the weighted representation shows more gradual decline as a function of cascade risk when compared to the binary representation. For the 11M network the decline is actually slightly more sharp in the weighted representation. Structurally, the 11M network has a center (measured by betweenness centrality) of tightly knit-nodes (very short distances), while the 9/11 network is more sparse at its center, increasing its cascade resilience. This effect explains the direction of the error in the binary representation. Based on those two examples, it appears that the binary representation does not have a systematic bias, and may even underestimate the fitness of dark networks.

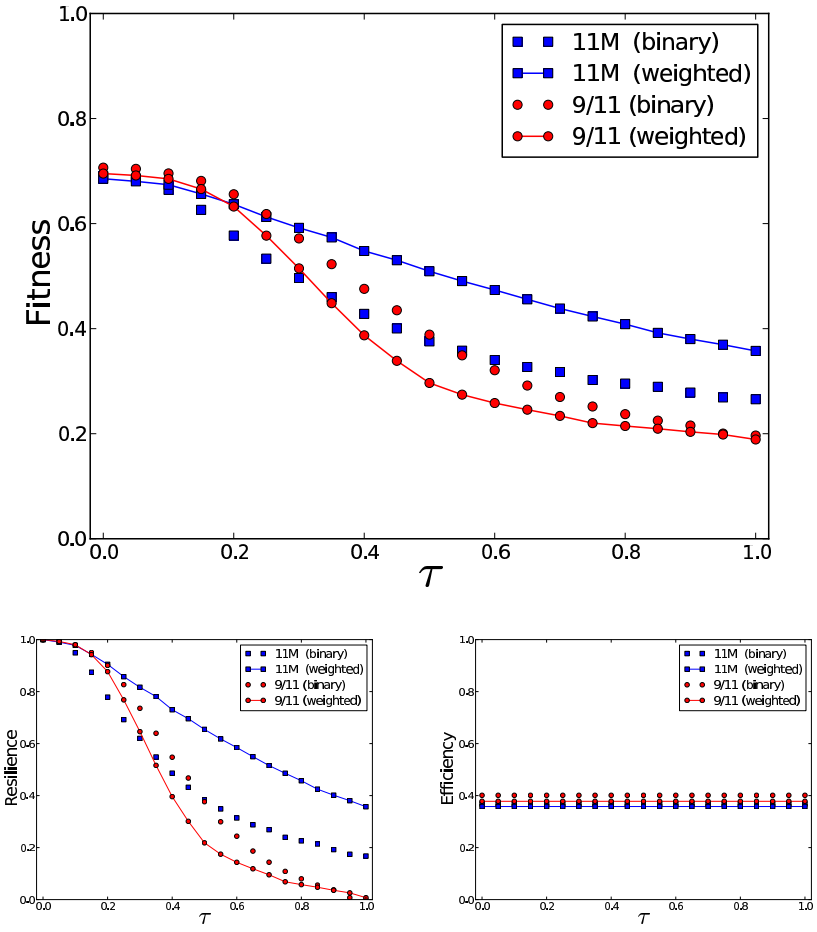


Fig. 7 Fitness, Resilience and Efficiency of two dark networks ($r = 0.51$), comparing binary and weighted representations. The binary representation matches the weighted representation within 0.15, and typically closer.

Appendix 2: Resilience and Efficiency of Empirical Networks

It is interesting to compare the empirical networks to each other in their efficiency and resilience (Fig. 8). Note that FTP and 9/11 networks are not the most resilient, but they strike a good balance between resilience and efficiency. The advantages of the two networks over other networks are not marginal, implying that their advantages in fitness are not sensitive to the choice of r. Of course, they are optimized for particular combinations of r and τ, and will no longer be very successful outside that range. For instance, in the range of high r and high τ networks with multiple

connected components would have higher fitness because they are able to isolate cascades in one component.

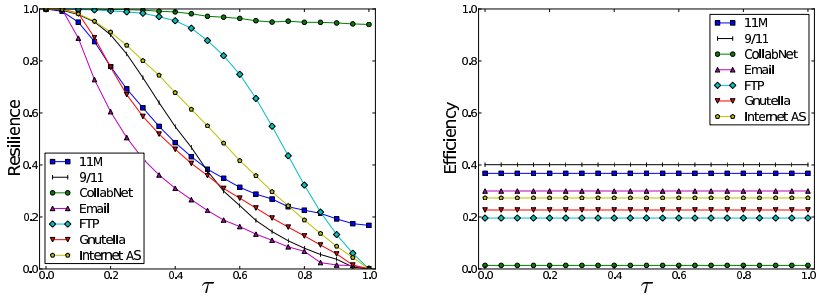


Fig. 8 Resilience and Efficiency of the real networks. The fittest networks are not always the most resilient.

Appendix 3: Network Designs

We considered networks on $n = 180$ nodes constructed through 6 simple designs, chosen both based on empirical findings (see e.g.[2, 13]) as well as the possibility of analytic tractability in some cases. When more data becomes available on dark networks, it will become possible to extract additional subgraphs with statistical validity.

Three of the designs are based on identical "cells": each cell is either (a) a clique (a complete graph), (b) a star (with a central node called "leader"), and (c) a cycle (nodes connected in a ring). Each of these have a single parameter, k - the number of nodes in the cell. Recent research suggests that under certain assumptions constructing networks from identical cells is optimal [27]. Let us also consider n-node graphs consisting of (d) randomly-connected cliques (sometimes termed "cavemen"), and (e) randomly-connected stars, in both cases according to probability p . Consider also (f) the simpler and well-studied Erdos-Renyi (ER) random graph with probability p (see figure in main text). By considering different structures for the cells we determine which of those structures provides the best performance.

The above palette of designs is sufficient for the current study's purpose of introducing an approach which could be applied to different domains, as well as begin constructing a theory to address cascade resilience. There are surely applications where if the objective is to design cascade-resilient networks, one ought to both reject some of the designs above for application-specific reasons and also to introduce additional designs. The task of taking other designs can be done by merely changing the computer program for generating the networks. Fruitful future research in cascade-resilient network is to attack the most general problem of finding the best

network on n nodes. Unfortunately, this search space is exponential and probably non-convex.

Research on social networks indicates that resilience and efficiency might be just two of several design criteria that also include e.g. "information-processing requirements", that impose additional constraints on network designs [5]. In the original context "information-processing" refers to the need to have ties between individuals involved in a particular task, when the task has high complexity. Each individual might have a unique set of expertise into which all the other agents must tap directly. Generalizing from sociology, such "functional constraints" might considerably limit the flexibility in constructing resilient and efficient networks. Such functional constraints could be addressed by looking at a palette of network designs which already incorporate such constraints or using innovative techniques from convex optimization [39].

The solution to the optimization problem is found by setting each of the parameters k (and when possible p) to various values. Each design D has "configurations" $C_1^D, C_2^D, \ldots$ each specifying the values of the parameters. Each configuration C_i^D is inputted to a program that generates an ensemble of $1-10$ networks, whose average performance provides an estimate of the fitness of C_i^D. The number of networks was 10 for networks with parameter p because there is higher variability between instances. The coefficient of variation (CV) in the fitness of the sample networks was monitored to ensure that the average is a reliable measure of performance. Typically CV was < 0.2 except near phase transitions of connectivity and percolation.

Optimization was performed using grid search. Alternative methods (e.g. Nelder-Mead) were considered but grid search was chosen despite its computational cost because it suffers no convergence problems even in the presence of noise (present due to variations in topology and contagion extent), and collects data useful for sensitivity analysis and multi-objective optimization. The sampling grid was as follows. In designs consisting of cells of size k, cell size was set to all integer values in $[1,180]$. If k did not divide 180, a cell of size $< k$ was added to ensure that the number of nodes in the graph is 180. The number of nodes is 180 because 180 is a highly-composite number and so it offers many networks of equally-sized cells. In general, normalization by n in the definitions of resilience and efficiency ensures that even when the number of nodes is tripled the effect of network size on fitness is very small for the above designs (around ± 0.05 in numerical experiments). In designs containing a parameter of connectivity p, it was set to all multiples of 0.05 in $[0,1]$, with some extra points added to better sample phase transitions. The grid search algorithm results are readily used to compute the Pareto frontier using the ε-balls method [46] ($\varepsilon = 0.01$).

The resilience metric is most easily computed by simulation where a node is selected at random to be "infected", and the simulation is run until all nodes are in states S or R, and none is in state I. In the simplest version of the SIR cascade model, which we adopt, each node in the graph can be in one of three states "susceptible", "infected" and "removed" designated S, I and R respectively (these names are borrowed from Epidemiology). Time is described in uniform discrete steps. A node in S state at time t stays in this state, unless a neighbor "infects" the node, causing it

to move to state I at time $t+1$. Specifically, a node in state S at time t has probability τ of turning to I state at time $t+1$ for each adjacent node in state I at time t. Finally, a node in I state at time t always becomes R at time $t+1$. Once in state R, the node remains there for all future times. It is possible to consider an alternate model where the rate of transition $I \to R$ takes more than one time step, but adding this effect would mostly serve to increase the probability of transmission, which is already parametrized by τ [58, 62].

A cascade/contagion that starts at a single node would run for up to n steps, but usually much fewer since typically $\tau < 1$ and/or the graph is not connected. To achieve good estimate of the average extent, the procedure was replicated 40 times, and then continued as long as necessary to achieve an error of under ± 0.5node with a 95% confidence interval [47].

An analytic computation of the cascade extent metric was investigated. It is possible in theory because the contagion is a Markov process with states in the superset of the set of nodes, 3^n. Unfortunately, such a state space is impractically large. When G is a tree, then an analytic expression exists[1], and it might be feasible when the treewidth is small [17, 58]. However, for many graph designs the tree approximation is not suitable. Another possible approach is to represent the contagion approximately as a system of differential equations which can be integrated numerically [40] . These possibilities were not pursued since the simulation approach could be applied to all graphs, while the errors of the analytic approaches are possibly quite large.

Appendix 4: Continuity of Fitness in r

We now justify the claim that fitness is continuous in r. In fact, we will prove the stronger property of Lipschitz-continuity. Notice that the claim is not about the continuity of fitness of a single configuration as a function of r but rather that:

Proposition 1. *$f(r) = \max_{G \in \mathbb{G}} F(G,r)$ is Lipschitz-continuous for $r \in [0,1]$.*

Proof. The argument constructs a bound on the change in f in terms of the change in r. Consider an optimal configuration C_1 of a design for $r = r_1$ and let its fitness be $f_1 = F(C_1, r_1)$ (there is slight abuse of notation since C is a configuration, whose fitness is the average fitness of an ensemble of graphs).
Observation 1: consider the fitness of C_1 at $r = r_2$. Because C_1 is fixed and the metrics are bounded ($0 \le R \le 1$ and $0 \le W \le 1$), the fitness change is bounded by the change in r:

[1] Specifically, the mean contagion size is $1 + \frac{pG_0'(1)}{1-pG_1'(1)}$, where $G_0(x)$ generates the degree distribution and $G_1(x) = \frac{G_0'(x)}{G_0'(1)}$ generates the probability of arrival to a node [58].

$$\begin{aligned} |f_1 - F(C_1, r_2)| &= |r_1 R(C_1) + (1-r_1)W(C_1) \\ &\quad -r_2 R(C_1) - (1-r_2)W(C_1)| \\ &= |(r_1 - r_2)R(C_1) - (r_1 - r_2)W(C_1)| \\ &\leq |r_1 - r_2| \,. \end{aligned}$$

Observation 2: let C_2 be the optimal configuration for $r = r_2$ and let $f_2 = F(C_2, r_2)$. Since C_2 is optimal for $r = r_2$ it satisfies: $f_2 \geq f(C_1, r_2)$, and so $-f_2 \leq -F(C_1, r_2)$. It follows that $f_1 - f_2 \leq f_1 - F(C_1, r_2)$. Take the absolute value of the right hand side and apply Observation 1 to get the bound: $f_1 - f_2 \leq |r_1 - r_2|$.
Observation 3: applying the argument of Observations 1&2 but reversing the roles of C_1 and C_2 implies that $f_2 - f_1 \leq |r_1 - r_2|$.

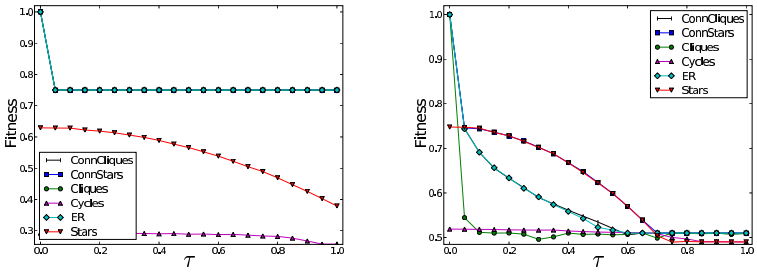
Observations 2&3 give $|f_1 - f_2| \leq |r_1 - r_2|$, proving the result.

Appendix 5: Additional Performance Results

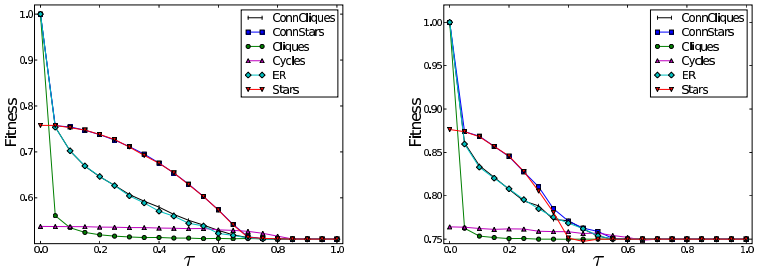
A plot of the fitness of various designs is in Fig. 9. Notice that for $r = 0.25$ Cliques outperform Stars. The resilience and efficiency are in Fig. 10 and 11.

The success of the Stars design could be analyzed more qualitatively. The fitness function combines resilience $R(G)$ which decreases when the graph becomes more strongly connected, and efficiency $W(G)$ which decreases when the graph becomes more sparse. The optimality of the star-based designs is due to a good trade-off between R and W: the central node in each cell (its "leader") provides a good firewall against cascades because in each cell most pairs are separated by distance of 2, but this separation reduces efficiency only modestly. In the Cliques design the separation is 1 (too short for resilience), and in the Cycles design it is too long ($\sim$a quarter of cell size k).

Mathematically, the existence of a non-trivial solution is due to the different functional relationships. To first-order approximation, efficiency decreases inversely with average distance,$\Theta\left(\frac{1}{\text{avg distance}^g}\right)$ while cascade extent decreases exponentially, $\Theta\left(\tau^{\text{avg distance}}\right)$, for $\tau < 1$ assuming a bounded number of alternative paths). For example, for the star design $R = O(1-\tau^2)$ and $W = O(2^{-g})$ as $n = k \to \infty$. Therefore, the optimal network's structure exploits the exponential decrease in cascades without sacrificing too much efficiency. In the range $\tau \in [0.2, 0.7]$ and $r \approx 0.5$, an average distance of ≈ 2, as in the star graph, might be optimal (cf. [50]).

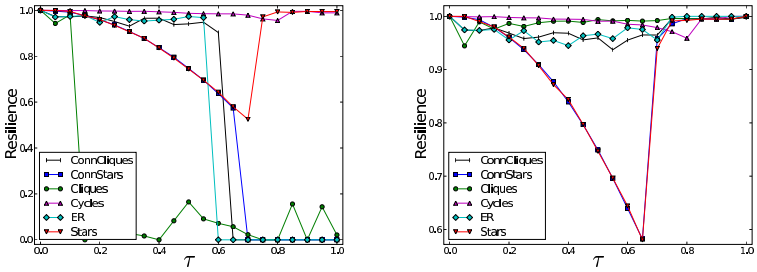


(a) $r = 0.25$ (b) $r = 0.49$



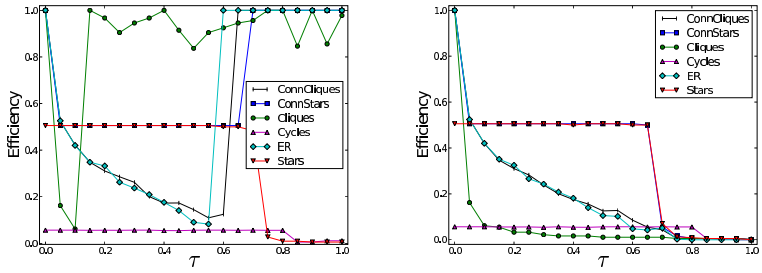
(c) $r = 0.51$ (d) $r = 0.75$

Fig. 9 Fitness of the optimal configuration. For very low r, such as $r = 0.25$ Cliques achieve higher fitness than simple Stars by forming the complete graph. Cliques never achieve higher fitness than the Connected Stars. For most values of r and τ, Stars are the best design.



(a) $r = 0.49$ (b) $r = 0.51$

Fig. 10 Resilience of the optimal design.



(a) $r = 0.49$ (b) $r = 0.51$

Fig. 11 Efficiency of the optimal design.

Appendix 6: Monotonicity of Fitness

Proposition 2. *Let $f(\tau)$ be the highest attainable fitness within a fixed network design $\mathbb{G}$, for cascade probability τ:*

$$f(\tau) = \max_{G \in \mathbb{G}} \left[\underbrace{rR(G,\tau) + (1-r)W(G)}_{F(G,\tau)} \right]$$

Then $f(\tau)$ is a non-increasing function of τ.

Proof. The proof relies on the simple claim that resilience of networks does not increase when τ increases [31]. The claim is equivalent to the result that for a given graph G increasing τ does not decrease the expected extent of cascades. The remainder is almost trivial: we need to show that when the fitness of all the points on the space (all graphs) has been made smaller or kept the same (by increasing τ), the new maximum value would not be greater than in the old space. Rigorously, assume by contradiction that $\tau_+ > \tau$ and fitness *increased*, namely*:*

$$f(\tau_+) > f(\tau)\,. \tag{6}$$

Let $G^*_{\tau_+}, G^*_\tau$ by any two optimal networks for τ_+ and τ, respectively, namely:

$$G^*_{\tau_+} \in \operatorname*{argmax}_{G \in \mathbb{G}} \left[rR(G,\tau_+) + (1-r)W(G) \right]$$
$$G^*_{\tau} \in \operatorname*{argmax}_{G \in \mathbb{G}} \left[rR(G,\tau) + (1-r)W(G) \right].$$

By optimality of G^*_τ at τ get that it must be at least as good at τ as $G^*_{\tau_+}$:

$$\underbrace{F(G^*_\tau,\tau) - F(G^*_{\tau_+},\tau)}_{\equiv \Delta} \geq 0\,. \tag{7}$$

The claim implies that:

$$R(G^*_{\tau_+}, \tau) \geq R(G^*_{\tau_+}, \tau_+)\,. \tag{8}$$

Expanding Δ:

$$\begin{aligned}
\Delta &= F(G^*_\tau, \tau) - \left[rR(G^*_{\tau_+}, \tau) + (1-r)W(G^*_{\tau_+})\right] \\
&\leq F(G^*_\tau, \tau) - rR(G^*_{\tau_+}, \tau_+) - (1-r)W(G^*_{\tau_+}) \text{ by (8)} \\
&= F(G^*_\tau, \tau) - F(G^*_{\tau_+}, \tau_+) \\
&< 0 \text{ by the hypothesis (6).}
\end{aligned}$$

This contradicts Ineq. 7.

The argument is easy to generalize. One could apply this method to the parameter g of attenuation, showing that fitness is non-increasing when attenuation is increased.

Appendix 7: Analytic Results

It is easy to analytically derive the values of the resilience, efficiency (and hence fitness) functions for certain simple designs: the Cycles and the Stars designs. These are useful to gain deeper insight into the effect of parameters. Recall that n is the number of nodes and k is the number of nodes per cell. For $k = 1$, in both designs $R = 1$ and $W = 0$. When $k \geq 2$, it is easy to verify that for the Cycle design:

$$R(n,k,\tau) = 1 - \frac{1}{n-1}\left[2\tau\frac{1-\tau^{k-1}}{1-\tau} - (k-1)\tau^k\right]$$

$$W(n,k,g) = \frac{1}{n-1}\begin{cases} \frac{1}{(\frac{k}{2})^g} + 2\sum_{j=1}^{\frac{k}{2}-1}\frac{1}{j^g} & \text{k even} \\ 2\sum_{j=1}^{\frac{k-1}{2}}\frac{1}{j^g} & \text{k odd} \end{cases}$$

and for the Stars design:

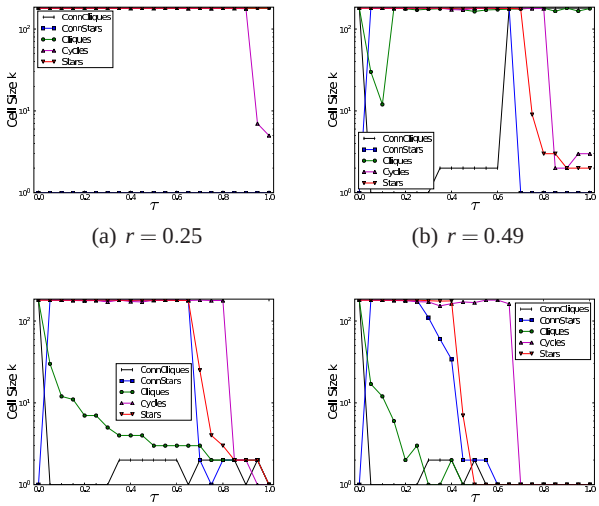
$$R(n,k,\tau) = 1 - \frac{1-\frac{1}{k}}{n-1}\left[2+\tau(k-2)\right]\tau$$

$$W(n,k,g) = \frac{1-\frac{1}{k}}{n-1}\left[2+2^{-g}(k-2)\right]\,.$$

These expressions are not readily useful for continuous optimization since k is discrete, but they can be used to identify phase transitions. Thus, they help inform optimization for designs where no analytic expression is available. The ER and Cliques designs are also analytically tractable (see e.g. [24] for expected cascade extent).

In the Stars design, when R and W are weighted equally ($r = \frac{1}{2}$), fitness takes a relatively simple form: $F = \frac{1}{2} + \frac{1}{2}\frac{1-\frac{1}{k}}{n-1}\left[2(1-\tau) + (2^{-g} - \tau^2)(k-2)\right]$. This implies

that increasing cell size k, for k large, improves fitness iff $2^{-g} - \tau^2 > 0$. Hence the optimal configuration has one cell ($k = n$), until a threshold near $\tau = 2^{-g/2}$ (for $g = 1$, approximately 0.71). This agrees with the findings in Fig. 12. Also, the rate of change in fitness with respect to τ, $\frac{dF}{d\tau} = \frac{1-\frac{1}{k}}{n-1}\left[-2-2\tau(k-2)\right]$, is always negative, as expected on more general grounds (see Appendix 6). It is linear in τ (because it is a tree graph) but superlinear in k (because of the mutual hazard induced by adding nodes to cells.)



(a) $r = 0.25$ (b) $r = 0.49$

(c) $r = 0.51$ (d) $r = 0.75$

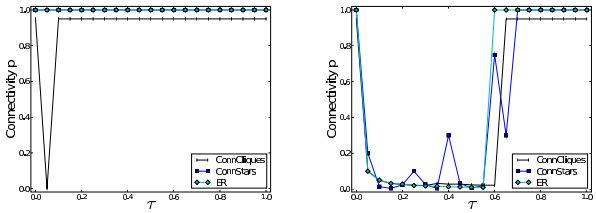
Fig. 12 Cell size k in the optimal configuration of each design. Cell size makes many sharp transitions.

Appendix 8: Configuration of the Optimal Design

As τ is varied, the optimal configuration changes. This section shows the resulting changes in the values of the parameters k (cell size) and p (connectivity). In other words, it indicates how each of the designs ought to be configured to attain optimal fitness, as a function of resilience weighting, r, and cascade probability, τ.

The cell size parameter k is non-monotonic for various designs when $r = 0.49$ (Fig. 12). For example, for the Connected Cliques design, at low contagion risk ($\tau < 0.1$), k is high (comparable to the size of the network, i.e. $k \to n$), then it falls to a small number. At high contagion risk ($\tau > 0.6$) the network is again highly connected again with $k \to n$. Thus for $\tau \to 1$, the optimal network is the fully-connected graph.

In general, designs involving both the p and k parameters show an interplay between the two (Fig. 13). For example, in the Connected Stars design under $r < 0.5$ there are two phase-transitions in connectivity p: as τ increases at $\tau \to \tau_l^* \approx 0.1$ it transitions from a connected graph to disconnected cells, and at $\tau \to \tau_u^* \approx 0.7$ back to full connectivity. If $r > 0.5$ the second transition is extinguished. The data requires care to interpret. For example, in the Connected Stars design $\tau \in [0.1, 0.65]$, when $r > 0.5$ the fluctuations in the p are noise because there is a single cell and a single cell leader ($k = n$), and so the parameter p has no effect. For sensitivity analysis see Appendix 9.



(a) $r = 0.25$ (b) $r = 0.49$

(c) $r = 0.51$ (d) $r = 0.75$

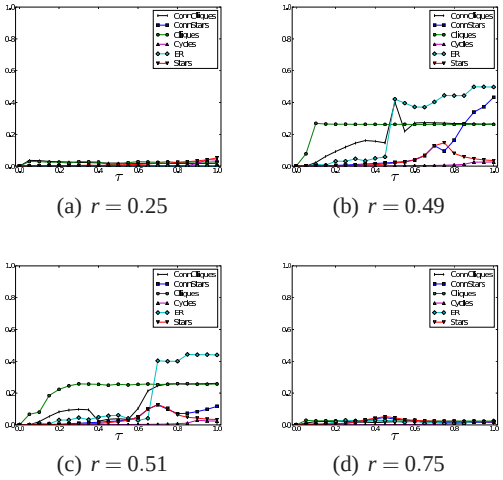
Fig. 13 Connectivity p in the optimal configuration of each design. Non-monotonic trends reflect structural transitions and sometimes also interplay with cell size.

Appendix 9: Sensitivity Analysis

It is desirable to determine how much variability exists within the optimal parameter values. Only if the variability is low the optimal configuration provides valuable information for network design. As proxy of robustness of the estimates, consider the space of configurations whose fitness ≥ 0.95 of the fitness of the optimal solution. The robustness of a parameter is measured by the parameter's standard deviation within this space (the unbiased estimate).

The results are in Figs. 14,15,17,16. Overall, as one would expect, the properties are more variable near the transition point $r = 0.5$, as compared to r values away

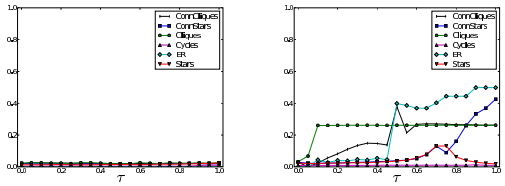
from $r = 0.5$. Moreover, variability is high within each design whenever the design undergoes a phase transition, since multiple different phases have nearly equal fitness. Another source of variance is found in designs with two parameters. These are more variable than designs with a single parameter because the former can reproduce some of the same graphs with many different parameter settings - the parameters have non-orthogonal effects. Finally, designs with low fitness for all configurations (like Cycles) show high variability since all configurations show uniform low performance.



(a) $r = 0.25$ (b) $r = 0.49$

(c) $r = 0.51$ (d) $r = 0.75$

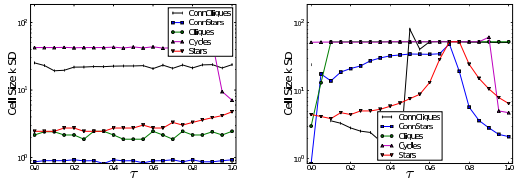
Fig. 14 Standard deviation in resilience within the top 5% of solutions.



(a) $r = 0.25$ (b) $r = 0.49$

(c) $r = 0.51$ (d) $r = 0.75$

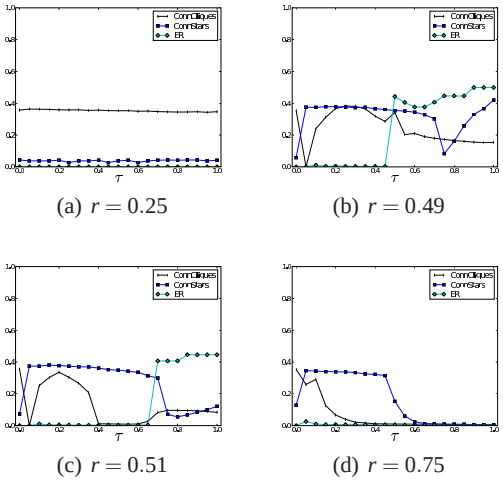
Fig. 15 Standard deviation in efficiency within the top 5% of solutions.



(a) $r = 0.25$ (b) $r = 0.49$

(c) $r = 0.51$ (d) $r = 0.75$

Fig. 16 Standard deviation in cell size k of the top 5% of solutions.



(a) $r = 0.25$ (b) $r = 0.49$

(c) $r = 0.51$ (d) $r = 0.75$

Fig. 17 Standard deviation in connectivity p of the top 5% of solutions.

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